
Where Classical Interpolation and a Convolutional Surrogate Fail: Wavefunction Super-Resolution on the 1D Quantum Harmonic Oscillator

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Abstract

1 We train a 1D convolutional network to reconstruct 1024-point ground-state wave-
2 functions of the quantum harmonic oscillator from 64-point coarse solutions, and
3 compare it with cubic-spline interpolation on 200 matched held-out samples. Both
4 clear the $F > 0.99$ fidelity threshold on every sample (network mean 0.999832,
5 spline 0.999790), so the comparison turns on how each method’s error is organized
6 over the frequency range. The difference is sharpest where the physics is easiest:
7 in the lowest-frequency bin the spline attains its lowest error while the network at-
8 tains its highest, on identical inputs. Spline infidelity is perfectly rank-correlated
9 with frequency (Spearman $\rho = +1.000$), a family-agnostic failure, whereas net-
10 work infidelity follows an inverted-U over the sampled interval, worst at both
11 edges ($\rho = +0.861$ against distance from the midpoint). Retraining on a wider
12 interval moved error off the old edges but not symmetrically onto the new ones,
13 only partly supporting a training-density explanation.

14 1 Introduction

15 Direct solution of the time-independent Schrödinger equation (TISE) on a fine grid scales badly:
16 an N -point 1D problem requires diagonalizing an $N \times N$ Hamiltonian, and dimension and particle
17 number compound the cost exponentially. A natural mitigation is to solve cheaply on a coarse
18 grid and reconstruct the fine solution with a learned model. We train a convolutional network,
19 QuantumResNet, to map a 64-point coarse ground state of the quantum harmonic oscillator to its
20 1024-point counterpart, scored by quantum fidelity.

21 The 1D oscillator does not need machine learning; it solves in tens of milliseconds and offers exact
22 ground truth across a densely sampled family. Since a cubic spline clears $F > 0.99$ across the range,
23 a higher mean fidelity would say little; the informative question is what organizes each method’s
24 error: the spline’s by resolution, the network’s by position within its training range. The model
25 is a supervised surrogate, not a physics-informed neural network: its loss is pure mean-squared
26 error against precomputed eigenvectors, with no Schrödinger-residual term, the defining feature of a
27 PINN (Raissi et al., 2019). “Quantum” here means quantum simulation on classical hardware, with
28 no qubits or circuits.

29 2 Related Work

30 PINNs embed a PDE residual in the training loss (Raissi et al., 2019); ours does not. Neural op-
31 erators such as the FNO learn resolution-independent maps with zero-shot super-resolution (Li et

al., 2021); ours is resolution-specific but reconstructs fine fields from coarse observations of a family indexed by ω , as in SRCNN (Dong et al., 2016). Closest are convolutional surrogates for the Schrödinger equation: Mills et al. (2017) map a 2D potential to a ground-state energy, whereas we reconstruct the eigenvector; Li et al. (2025) apply super-resolution-inspired upsampling to electron densities, scoring by energy error. Neural-network quantum states parameterize the wavefunction variationally (Carleo and Troyer, 2017). That learned surrogates degrade away from their training distribution is well documented: Mouli et al. (2024) report neural operators failing on out-of-domain PDE inputs at high in-domain accuracy, and Gopakumar et al. (2026) give conformal coverage across a surrogate’s operating domain under an exchangeability assumption. What this setting adds is separability: a classical baseline on identical inputs has error ordered by physics without exception ($\rho = +1.000$, zero inversions across 200 samples), so both failure modes are observable side by side.

3 Methods

System and ground truth. We solve the 1D TISE in atomic units ($\hbar = m = 1$), $-\frac{1}{2}\psi'' + V(x)\psi = E\psi$, with $V(x) = \frac{1}{2}\omega^2 x^2$ on $[-10, 10]$ with Dirichlet boundaries. A three-point finite-difference stencil yields a sparse Hamiltonian; the ground state comes from `eigsh(H, k=1, which="SA")` at $N = 1024$ (target) and $N = 64$ (input). Wavefunctions are L^2 -normalized with fixed sign; only the nodeless $n = 0$ state is used, at fixed centre $x_0 = 0$. The state is a Gaussian of width $\sigma = 1/\sqrt{\omega}$ against spacing $\Delta x = 20/63 \approx 0.317$, so coarse points per width fall from 4.45 at $\omega = 0.5$ to 1.41 at $\omega = 5.0$.

Data and training. 2,000 samples, $\omega \sim U[0.5, 5.0]$, seed 42; split 72/18/10 (1,440/360/200). QuantumResNet maps $[B, 1, 64] \rightarrow [B, 1, 1024]$ through four transposed-convolution stages (kernel 4, stride 2, padding 1), each followed by a pre-activation residual block (BatchNorm/GELU/Conv1d, twice, with skip); 64 channels, 166,209 parameters. Adam (lr 10^{-3} , batch 64), MSE loss, cosine annealing over the 30-epoch budget; early stopping (patience 7) restored the best-validation checkpoint. A run takes 113 s on an Intel Core Ultra 7 256V (CPU). Early stopping did not fire, which is not headroom: the schedule drives the learning rate to near zero by epoch 30. Robustness checks retrain the same configuration under seeds 43–46; one single-seed ablation uses $\omega \in [0.3, 7.0]$, $N = 3,000$ (448 samples per unit ω vs the canonical 444).

Metrics and baselines. Fidelity is $F = (\sum_i \psi_{\text{hr},i} \psi_{\text{pred},i} \Delta x)^2$, both renormalized under the released code’s discrete-sum rule; we report infidelity $1 - F$. Baselines: a cubic spline (`CubicSpline`, $64 \rightarrow 1024$); Fourier zero-padding applied to the 63 unique periodic points of the endpoint-inclusive grid; and two oracles that assume the harmonic family. The *moment oracle* estimates ω from the coarse input through $\text{var}(x) = 1/(2\omega)$ and emits the analytic state $(\omega/\pi)^{1/4} \exp(-\omega x^2/2)$. The *inversion oracle* recovers ω by minimizing $1 - F$ between the input and the coarse finite-difference state at ω (bounded Brent, `xatol` 10^{-8} , seeded from the moment estimate), then emits the fine-grid state there; costing a fine solve per sample, it is not a candidate method but a bound on the information the coarse input carries. None is tuned. Error shapes use Spearman correlations of infidelity against ω and against $|\omega - \omega_{\text{mid}}|$ ($\omega_{\text{mid}} = 2.750$); the second predictor is a folded transform of the first, so the pair restates Table 1 rather than testing between explanations.

4 Results

Fidelity. On the 200-sample test set the network reached mean fidelity 0.999832 (median 0.999889, min 0.999328, std 1.13×10^{-4}), 0/200 below $F = 0.99$; the spline reached 0.999790 (min 0.999442, std 1.65×10^{-4}). Across seeds 42–46 network mean fidelity is 0.999869 (0.999832–0.999914); seed 42, the headline run, is the lowest. The Fourier baseline initially scored 0.969008 through a grid-convention error, not a property of Fourier interpolation: an endpoint-inclusive array transformed as periodic in index space, shifting it 0.149 units. Corrected it scores 0.999770. The moment oracle reaches 0.999789 and tracks the spline bin by bin rather than the network, its ω estimate carrying a mean $|\Delta\omega|$ of 0.13. The inversion oracle recovers ω to a mean $|\Delta\omega|$ of 6.09×10^{-8} (max 2.81×10^{-7}) and matches the target to a mean infidelity below 1.1×10^{-16} in every bin, at double-precision round-off. The coarse input therefore determines the state completely within the family, so the spline’s losses at high ω reflect family-agnosticism and the moment oracle’s a mis-specified

Table 1: Fidelity and infidelity by ω bin; all methods on the same 200 matched samples (seed 42). FFT is the corrected Fourier baseline.

ω range	n	Spline F	FFT F	Moment F	CNN F	Spline $1 - F$	CNN $1 - F$
[0.5, 1.4)	36	0.999980	0.999979	0.999981	0.999680	$2.03\text{e}-5$	$3.20\text{e}-4$
[1.4, 2.3)	36	0.999932	0.999930	0.999936	0.999894	$6.75\text{e}-5$	$1.06\text{e}-4$
[2.3, 3.2)	46	0.999836	0.999826	0.999840	0.999896	$1.64\text{e}-4$	$1.04\text{e}-4$
[3.2, 4.1)	42	0.999708	0.999682	0.999709	0.999892	$2.92\text{e}-4$	$1.08\text{e}-4$
[4.1, 5.0]	40	0.999527	0.999466	0.999511	0.999777	$4.73\text{e}-4$	$2.23\text{e}-4$
Overall	200	0.999790	0.999770	0.999789	0.999832	$2.10\text{e}-4$	$1.68\text{e}-4$

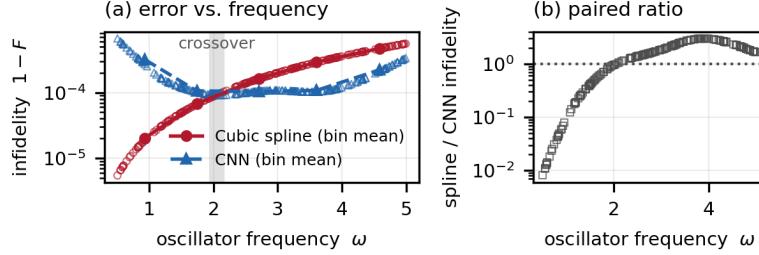


Figure 1: (a) Infidelity against ω for spline and network on the 200 matched samples (open markers) with bin means, log ordinate; the curves cross near $\omega \approx 2$ (shaded). (b) Per-sample ratio of spline to network infidelity, crossing unity at the same transition.

estimator, the analytic identity applied to a finite-difference state, rather than missing information; the network’s 1.68×10^{-4} lies far above what the input permits.

Matched comparison. The network wins 132 of 200 matched samples (66%; one-sided binomial $p = 3.5 \times 10^{-6}$; Wilcoxon on paired infidelities $W = 6588$, $p = 2.4 \times 10^{-5}$), with $1.25\times$ lower mean infidelity. Both clear $F > 0.99$ on every sample, so these are differences in the structure of very small errors, not a pass/fail separation. The aggregate conceals a reversal (Figure 1): the network loses all 36 samples in the lowest bin and 32 of 36 in the second, then wins all 128 above; below $\omega = 1.5$ it wins 0 of 42, and at $\omega \geq 3.5$ 65 of 65 with $2.31\times$ lower mean infidelity. For this seed the spline’s last win falls at $\omega = 2.03$ and the network wins every sample from $\omega = 2.09$ up; across five seeds that run begins between $\omega = 1.73$ and 2.10 (mean 1.95) with 122–157 wins of 200, so the crossover is a band, not a fixed value.

Two error structures. The clearest evidence is where the physics is easiest. In the lowest bin the spline attains 2.03×10^{-5} , the lowest error of any method in any bin, while the network attains 3.20×10^{-4} , its own worst and three times its interior error. All three non-learned baselines attain their lowest error in that bin, where the network attains its highest, so undersampling cannot be what the network is failing on. Numerically, spline infidelity is perfectly rank-correlated with ω ($\rho = +1.000$, zero inversions) and uncorrelated with the folded predictor ($\rho = +0.031$, $p = 0.66$); network infidelity is uncorrelated with ω ($\rho = +0.026$, $p = 0.71$) and strongly correlated with distance from the midpoint ($\rho = +0.861$, $p = 4.0 \times 10^{-60}$). Re-expressing the predictor as $\sigma = 1/\sqrt{\omega}$ or $\log \omega$ leaves these unchanged up to sign, as monotone reparametrization must, excluding a parametrization artefact. Low-edge infidelity exceeds high-edge by $1.433\times$.

A distributional reading fits this: accuracy set by local training density, worst where a sample has neighbours on one side only. Retraining once on $\omega \in [0.3, 7.0]$ at matched density held that prediction only in part. Both former edges improved once interior, but unequally: the high edge fell from 2.23×10^{-4} to 5.45×10^{-5} , indistinguishable from the interior $5\text{--}6 \times 10^{-5}$, while the low edge improved about threefold, from 3.20×10^{-4} to 1.09×10^{-4} , but remains roughly twice interior. Error then concentrated at the new low edge (3.15×10^{-4} for $\omega < 0.5$, $n = 9$) but only mildly at the new high edge (7.40×10^{-5}); mean fidelity rose to 0.999923 and the crossover moved to $\omega = 1.53$. Edge error follows the training boundary, but asymmetrically, which distance to the boundary alone does not explain, and the alternative that extremal widths are intrinsically hard for a fixed receptive field is not excluded. The ablation is one seed, its new low bin nine samples.

5 Discussion

The spline’s error is bounded not by its input’s information but by family-agnosticism: the inversion oracle recovers ω from the same 64 samples to 6.09×10^{-8} , so the information is present and a family-blind interpolant cannot use it. That gap also bears on the network, which reaches 1.68×10^{-4} where inverting the same family reaches round-off; inversion costs a fine solve per sample and is no competing method, but bounds what the network leaves unclaimed. Its error is instead a function of training-set design: moving a boundary moves the error, though not symmetrically. A candidate explanation is that the relevant density is not in ω but in the shape reproduced, since uniform sampling in ω gives density per unit width σ scaling as $2/\sigma^3$, leaving broad, low- ω states sparsest even after widening. That is an inference, not a tested claim; a direct test would sample uniformly in σ or $\log \omega$.

A corollary cuts against the network: a surrogate whose accuracy tracks its training data is weakest where this class of method is meant to be used, where ground truth is scarcest (Mouli et al., 2024; Gopakumar et al., 2026); this setting isolates that, since a physics-ordered control on identical inputs makes the departure observable.

6 Limitations

Scope. Every sample is a harmonic ground state at fixed $x_0 = 0$, only ω randomized; shifted potentials, other forms (double-well and anharmonic, implemented but untrained), and excited states are untested. Test ω lies inside the training range, so this characterizes degradation toward the edges, not extrapolation.

Mechanism attribution. The distributional explanation rests on the low- ω asymmetry, the error shape, and one widened-range retrain. That retrain moved edge error inward but left the new edges unequal, settling neither the distance-to-boundary account nor the receptive-field alternative, and raising a third, untested candidate: sampling density in shape space. It is single-seed, its new low bin holds nine samples, and trains on 1.5 times the samples and gradient steps of the canonical run, so its lower interior error is not purely a range effect. The rank correlations add no independent support, and “interior” is unambiguous only in one dimension.

Seed variance. Headline numbers come from one model and one split (seed 42). Across five seeds, network mean fidelity spans 0.999832 to 0.999914 and the crossover $\omega = 1.73$ to 2.10, so the Section 4 crossover values are one realization and the margin varies with seed; the inverted-U with a worse low edge appeared in every seed. The architecture was fixed a priori, not ablated.

Fidelity is the only accuracy metric. It is an amplitude-weighted global overlap, insensitive to high-frequency content and boundary violation, both visible in this model’s residuals. Since the kinetic operator involves a second derivative, a reconstruction scoring well on fidelity may score worse on energy or $\|H\psi - E\psi\|$; no such metric was computed, and the ranking may not survive one.

7 Conclusion

The two methods’ errors are organized by different variables, most visibly where the physics is easiest. Spline error is perfectly rank-ordered by ω ; since inverting the coarse eigenproblem recovers ω to 6.09×10^{-8} , the input determines the state within the family, so that failure is representational, not informational. Network error follows an inverted-U that five seeds reproduce, and widening the range moved error off the old edges but not symmetrically onto the new ones, leaving training density in shape space as the open question. Next steps: training uniform in σ or $\log \omega$; physics-based evaluation with energy and residual metrics; and extension to the double-well and anharmonic potentials.

Use of LLMs. LLM assistance was used for drafting, analysis code (Fourier correction, both oracles, ablation, seed replication) and auditing of numbers and citations; all results were verified against the released code. **Code and data.** An anonymized snapshot of the code, checkpoints, metrics and analysis scripts accompanies this submission and regenerates every number reported.

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